

Practical Algorithms and Fixed Parameter Tractability for the Single Individual SNP Haplotyping Problem

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The Problem

■ A SNP Matrix.

	1	2	3	4	5	6
1	1	0	-	1	1	0
2	1	-	-	1	1	-
3	0	1	-	-	0	-
4	-	1	0	1	0	1
5	-	1	0	-	0	1
6	0	-	0	0	0	-

■ MSR (Minimum SNP Removal)

	1	2	3	5	6
1	1	0	-	1	0
2	1	-	-	1	-
3	0	1	-	0	-
4	-	1	0	0	1
5	-	1	0	0	1
6	0	-	0	0	-

■ MFR (Minimum Fragment Removal)

	1	2	3	4	5	6
1	1	0	-	1	1	0
2	1	-	-	1	1	-
3	0	1	-	-	0	-
4	-	1	0	1	0	1
5	-	1	0	-	0	1

MSR (Gapless Case)

- SNP conflict

...	1	...	1	...
...	0	...	1	...

- We define $K[j]$ as the maximum number of columns that can be kept to make M error-free.

- A dynamic programming $O(mn^2)$ algorithm.

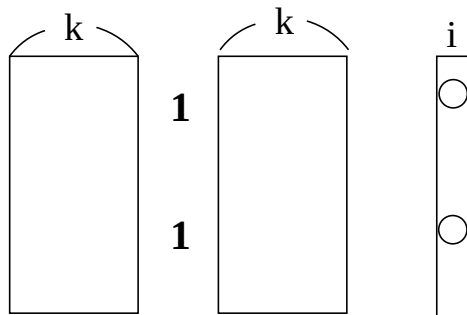
	1	2	3	4	5
1	0	1	0	1	1
2	0	1	0	-	-
3	1	0	1	0	-
4	-	0	1	0	0
5	-	1	1	1	-

MSR with Gaps

- An $O(mn^{2k+2})$ algorithm.

	1	2	3	4	5	6
1	1	0	1	-	1	0
2	1	-	-	1	1	-
3	0	1	-	-	0	-
4	-	1	-	1	0	1
5	-	1	0	0	0	1
6	0	-	0	0	0	-

- Considering $2k$ columns of history.



MFR (Gapless Case)

- Fragment conflict

$$\begin{array}{|c} 1 \\ 0 \end{array}$$

- We define $D[h, k; i]$ as the minimum number of rows to remove to make M_i error-free.
- A dynamic programming $O(m^2n + m^3)$ algorithm.

	1	2	3	4	5	6
1	1	0	1	0	1	1
2	-	0	1	0	-	-
3	-	1	0	1	0	-
4	-	-	1	1	0	0
5	-	-	0	1	0	-

MFR with Gaps

- An $O(2^{2k}nm^2 + 2^{3k}m^3)$ algorithm.

	1	2	3	4	5	6
1	1	0	1	-	1	0
2	1	-	-	1	1	-
3	0	1	-	-	0	-
4	-	1	-	1	0	1
5	-	1	0	0	0	1
6	-	-	0	0	0	-

	1	2	3	4	5	6
1	1	0	1	-	1	0
2	1	-	-	1	1	-
3	0	1	-	-	0	-
4	-	1	-	1	0	1
5	1	1	0	0	0	1
6	1	1	0	0	0	1

The Haplotyping Problem

- MFR. (Minimum Fragment Removal)
- MSR. (Minimum SNP Removal)
- LHR. (Longest Haplotype Reconstruction)
- MEC. (Minimum Error Correction)
- Known results.

Problems	Gaps	Exact	Approximation
MFR	≤ 1	NP-hard	APX-hard ¹
	0	Solvable in time $O(m^2n + m^3)$	
	k holes	Solvable in time $O(2^{2k}nm^2 + 2^{3k}m^3)$	
MSR	≤ 2	NP-hard	APX-hard ²
	0	Solvable in time $O(mn^2)$	
	k holes	Solvable in time $O(mn^{2k+2})$,	
LHR	k holes	?	?
	0	Polynomial time	
MEC	n	NP-hard	?
	0	?	

Table 1: Known results for some problem on fragments